POTNURU SAMITA

ECE Spl. in Artificial Intelligence and Machine Learning

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OBJECTIVE

Final-year B.Tech student with experience in React.js, JavaScript, frontend development, and full-stack web technologies. Passionate about building responsive, scalable, and user-centric web applications.

INTERNSHIPS

Research Intern (Summer Fellowship Programme) – IIT Madras 05/2025 – 07/2025 | Chennai
- Designed a streaming IP core to compute log-power, perform dynamic peak detection, and generate histograms for frequency analysis.
- Implemented and optimized the architecture using Vitis HLS and Vivado 2024 on an FPGA development board.
- Improved hardware resource utilization and processing efficiency through architecture-level optimizations.

Research Intern (SURE Programme) – IIT Hyderabad 05/2024 – 07/2024 | Hyderabad
- Implemented Python-based machine learning workflows using Google Colab for data processing and model experimentation.
- Performed data preprocessing, feature analysis, and model evaluation on research datasets.
- Studied BCZT ceramics, including synthesis techniques, electrical characteristics, and XRD analysis for material characterization.

EDUCATION

B.Tech - Gandhi Institute of Technology and Management (GITAM) University - CGPA: 9.01 / 10 2026

MPC - Sri Chaitanya College - CGPA: 8.73 / 10 2022

SSC - Amar School - CGPA: 9.0 / 10 2020

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, Java, HTML5, CSS3, C#
Frontend Development: React.js, TypeScript, React Hooks, Context API, Component-Based Architecture, Responsive Web Design, Cross-Browser Compatibility, REST API Integration, Frontend Performance Optimization, UI Development
UI Frameworks & Styling: Tailwind CSS, Material UI, CSS Flexbox, CSS Grid
Backend Development: Node.js, Express.js, .NET, RESTful APIs, CRUD Operations, Authentication & Authorization
Database Technologies: PostgreSQL, SQL, Database Design, Database Management
Development Tools: Git, GitHub, VS Code, Postman, npm, Vite, Chrome Developer Tools, Cloud, Data, ERP
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Version Control, Debugging
AI & Machine Learning: Deep Learning, Computer Vision, TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn, RAG, LLM, Supabase, Firebase, n8n, MCP, Windsurf, Codex, Claude

CERTIFICATIONS AND COMPETITIONS

CodeHer Hackathon - WLF 01/2025
- Built an attendance tracking solution
- Secured 5th place

ICTIEE 2025 - VNRVJIET, Hyderabad 01/2025
- Presented IoT-based Sustainable Bridge
- Recognized among top 20% of submissions

ACHIEVEMENTS

Academic Achievement - ECE Department
- Ranked among the top 20% academically.
- Received 2 times Merit Scholarship.

EXPERTISE

Problem Solving, Software Engineering, Team Collaboration, Communication, Leadership, Adaptability, Attention to Detail

PROJECTS

YatraAI - AI Powered Heritage Tourism Platform 2026
- Developed a full-stack heritage tourism platform using React.js, TypeScript, TailwindCSS, Flask, and CNN.
- Built responsive user interfaces featuring monument recognition, multilingual AI assistance, interactive maps, and travel planning tools.
- Integrated Firebase Authentication, REST APIs, Leaflet Maps, and weather services to deliver real-time user experiences.
- Implemented AI-powered monument identification using CNN models with Groq AI integration for travel guidance.

Daily Productivity Tracker - Personal Project 2026
- Developed a full-stack productivity tracking application using React.js, Spring Boot, JWT Authentication, and PostgreSQL.
- Built responsive dashboards for task management, productivity analytics, and efficiency tracking with CRUD and REST API integration.
- Implemented secure user authentication, task filtering, and real-time performance metrics using React components and Spring Security.

Real-Time Human Motion Classification Using ESP32 - IIT Bhubaneswar 03/2025 – 05/2025
- Developed and deployed an edge AI solution on ESP32 for real-time Parkinson's disease detection using IMU sensor data.
- Built machine learning pipelines using Python and Google Colab, achieving 85% classification accuracy through signal processing and model optimization.

POSITIONS OF RESPONSIBILITY

IETE Student Chapter — GITAM — Core Member 10/2024 – 03/2026

Entrepreneurs Club — GITAM — Coordinator 10/2023 – 12/2024